

Hands-on Assignments for Displaying Data from Multiple Tables Using Joins

Complete the below hands-on assignments before proceeding with the next Topic

Update Completion Status

No.	Cat	Hands-on Assignment	Topics Covered
1	M	Write a query for the HR department to produce the addresses of all the departments. Use the EMP and DEPT tables. Show the EMPNO, ENAME, SAL, DNAME and LOC in the output. Use a NATURAL JOIN to produce the results.	Natural Join

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

Worksheet

Query Builder

```

SELECT e.EMPLOYEE_ID AS EMPNO,
       e.LAST_NAME AS ENAME,
       e.SALARY AS SAL,
       d.DEPARTMENT_NAME AS DNAME,
       l.CITY AS LOC
FROM HR.EMPLOYEES e
NATURAL JOIN HR.DEPARTMENTS d
JOIN HR.LOCATIONS l ON d.LOCATION_ID = l.LOCATION_ID;

```

Script Output

Task completed in 0.191 seconds

EMPNO	ENAME	SAL	DNAME	LOC
101	Yang	17000	Executive	Seattle
102	Garcia	17000	Executive	Seattle
104	Miller	6000	IT	Southlake
105	Williams	4800	IT	Southlake
106	Jackson	4800	IT	Southlake
107	Nguyen	4200	IT	Southlake
109	Faviet	9000	Finance	Seattle
110	Chen	8200	Finance	Seattle
111	Sciarra	7700	Finance	Seattle
112	Urman	7800	Finance	Seattle
113	Popp	6900	Finance	Seattle
115	Khoo	3100	Purchasing	Seattle
116	Baida	2900	Purchasing	Seattle
117	Tobias	2800	Purchasing	Seattle
118	Himuro	2600	Purchasing	Seattle
119	Colmenares	2500	Purchasing	Seattle
129	Bissot	3300	Shipping	South San Francisco
130	Atkinson	2800	Shipping	South San Francisco
131	Marlow	2500	Shipping	South San Francisco
132	Olson	2100	Shipping	South San Francisco

2	M	The HR department needs a report of all employees. Write a query to display the JOB, MGR, SAL, COMM, DNAME of employees whose JOB is SALESMAN.	Equi Join
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Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

Query Builder

```

SELECT e.JOB_ID AS JOB,
       e.MANAGER_ID AS MGR,
       e.SALARY AS SAL,
       e.COMMISSION_PCT AS COMM,
       d.DEPARTMENT_NAME AS DNAME
FROM HR.EMPLOYEES e
JOIN HR.DEPARTMENTS d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
WHERE e.JOB_ID IN ('SA_MAN', 'SA_REP');

```

Script Output

Task completed in 0.635 seconds

JOB	MGR	SAL	COMM	DNAME
SA_MAN	100	14000	.4	Sales
SA_REP	149	6200	.1	Sales
SA_MAN	100	12000	.3	Sales
SA_MAN	100	11000	.3	Sales
SA_MAN	100	10500	.2	Sales
SA_REP	145	10000	.3	Sales
SA_REP	145	9500	.25	Sales
SA_REP	145	9000	.25	Sales
SA_REP	145	8000	.2	Sales
SA_REP	145	7500	.2	Sales
SA_REP	145	7000	.15	Sales
SA_REP	146	10000	.35	Sales
SA_REP	146	9500	.35	Sales
SA_REP	146	9000	.35	Sales
SA_REP	146	8000	.3	Sales
SA_REP	146	7500	.3	Sales
SA_REP	146	7000	.25	Sales
SA_REP	147	10500	.25	Sales
SA_REP	147	9500	.15	Sales
SA_REP	147	7200	.1	Sales

3

M

The HR department needs a report of employees in LOC DALLAS. Display the ENAME, job, DEPTNO, and DNAME for all employees who work in DALLAS.

Equi Join

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

Query Builder

```

SELECT e.LAST_NAME AS ENAME,
       e.JOB_ID AS JOB,
       e.DEPARTMENT_ID AS DEPTNO,
       d.DEPARTMENT_NAME AS DNAME
FROM HR.EMPLOYEES e
JOIN HR.DEPARTMENTS d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
JOIN HR.LOCATIONS l ON d.LOCATION_ID = l.LOCATION_ID
WHERE l.CITY = 'DALLAS';

```

Script Output

Task completed in 0.125 seconds

4

M

Create a report to display employees' ename and employee number along with their manager's name and manager number. Label the columns Employee, Emp#, Manager, and Mgr#, respectively.

Self Join

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet Query Builder

```
SELECT e.LAST_NAME AS "Employee",
       e.EMPLOYEE_ID AS "Emp#",
       m.LAST_NAME AS "Manager",
       m.EMPLOYEE_ID AS "Mgr#"
FROM HR.EMPLOYEES e
JOIN HR.EMPLOYEES m ON e.MANAGER_ID = m.EMPLOYEE_ID;
```

Script Output

Task completed in 0.147 seconds

Employee	Emp#	Manager	Mgr#
Ozer	168	Cambault	148
Bloom	169	Cambault	148
Fox	170	Cambault	148
Smith	171	Cambault	148
Bates	172	Cambault	148
Kumar	173	Cambault	148
Vishney	162	Errazuriz	147
Greene	163	Errazuriz	147
Marvins	164	Errazuriz	147
Lee	165	Errazuriz	147
Ande	166	Errazuriz	147
Employee	Emp#	Manager	Mgr#
Banda	167	Errazuriz	147
Bissot	129	Frapp	121
Atkinson	130	Frapp	121
Marlow	131	Frapp	121
Olson	132	Frapp	121
Sarchand	184	Frapp	121
Bull	185	Frapp	121
Dellinger	186	Frapp	121
Cabrio	187	Frapp	121

5

M

Modify the previous Query to display all employees including King, who has no manager. Order the results by the employee number.

Outer Join

Oracle SQL Developer : Oracle

Find Database Object

Oracle

- ☒ All Schemas
- ☒ All Object Types
- ☐ All Identifier Types
- ☐ All Identifier Usages
- ☐ Columns
- ☐ Code
- ☐ All Dependencies

Worksheet

Query Builder

```

SELECT e.LAST_NAME AS "Employee",
       e.EMPLOYEE_ID AS "Emp#",
       m.LAST_NAME AS "Manager",
       m.EMPLOYEE_ID AS "Mgr#"
FROM HR.EMPLOYEES e
LEFT OUTER JOIN HR.EMPLOYEES m ON e.MANAGER_ID = m.EMPLOYEE_ID
ORDER BY e.EMPLOYEE_ID;

```

Script Output

Task completed in 0.155 seconds

Employee	Emp#	Manager	Mgr#
King	100		
Yang	101	King	100
Garcia	102	King	100
James	103	Garcia	102
Miller	104	James	103
Williams	105	James	103
Jackson	106	James	103
Nguyen	107	James	103
Gruenberg	108	Yang	101
Faviet	109	Gruenberg	108
Chen	110	Gruenberg	108
Sciarra	111	Gruenberg	108
Urman	112	Gruenberg	108
Popp	113	Gruenberg	108
Li	114	King	100
Khoo	115	Li	114
Baida	116	Li	114
Tobias	117	Li	114
Himuro	118	Li	114
Colmenares	119	Li	114

6

M

The HR department needs a report on job grades and salaries. To familiarize yourself with the SALGRADE table, first show the structure of the SALGRADE table. Then create a query that displays the name, job, department name, salary, and grade for all employees.

Non-Equi Join

Find Database Object

Oracle

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

Oracle SQL Developer : Oracle

hr_install.sql

0.18099999 seconds

Worksheet

Query Builder

```

WITH SALGRADE AS (
  SELECT 1 AS GRADE, 0 AS LOSAL, 1000 AS HISAL FROM DUAL UNION ALL
  SELECT 2, 1001, 2000 FROM DUAL UNION ALL
  SELECT 3, 2001, 3000 FROM DUAL UNION ALL
  SELECT 4, 3001, 5000 FROM DUAL UNION ALL
  SELECT 5, 5001, 99999 FROM DUAL
)
SELECT e.LAST_NAME      AS NAME,
       e.JOB_ID        AS JOB,
       d.DEPARTMENT_NAME AS DNAME,
       e.SALARY        AS SALARY,
       sg.GRADE        AS GRADE
FROM   HR.EMPLOYEES e
JOIN   HR.DEPARTMENTS d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
JOIN   SALGRADE sg ON e.SALARY BETWEEN sg.LOSAL AND sg.HISAL;

```

Script Output

Task completed in 0.181 seconds

NAME	JOB	DNAME	SALARY	GRADE
Whalen	AD_ASST	Administration	4400	4
Davis	MK_REP	Marketing	6000	5
Martinez	MK_MAN	Marketing	13000	5
Colmenares	PU_CLERK	Purchasing	2500	3
Himuro	PU_CLERK	Purchasing	2600	3
Tobias	PU_CLERK	Purchasing	2800	3
Baida	PU_CLERK	Purchasing	2900	3
Khoo	PU_CLERK	Purchasing	3100	4
Li	PU_MAN	Purchasing	11000	5
Jacobs	HR_REP	Human Resources	6500	5
Olson	ST_CLERK	Shipping	2100	3
NAME	JOB	DNAME	SALARY	GRADE
Markle	ST_CLERK	Shipping	2200	3
Philtanker	ST_CLERK	Shipping	2200	3

7

M

Display the ENAME,DNAME of all the employees. Also display those department name which do not have any employees working.

Outer Join

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

☒ All Schemas
☒ All Object Types
☐ All Identifier Types
☐ All Identifier Usages
☐ Columns
☐ Code
☐ All Dependencies

hr_install.sql

0.147 seconds

Worksheet

```

SELECT e.LAST_NAME AS ENAME,
       d.DEPARTMENT_NAME AS DNAME
FROM HR.EMPLOYEES e
FULL OUTER JOIN HR.DEPARTMENTS d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID;

```

Script Output

Task completed in 0.147 seconds

ENAME	DNAME
King	Executive
Yang	Executive
Garcia	Executive
James	IT
Miller	IT
Williams	IT
Jackson	IT
Nguyen	IT
Gruenberg	Finance
Faviet	Finance
Chen	Finance

Sciarra	Finance
Urman	Finance
Popp	Finance
Li	Purchasing
Khoo	Purchasing
Baida	Purchasing
Tobias	Purchasing
Himuro	Purchasing
Colmenares	Purchasing
Weiss	Shipping
Fripp	Shipping
ENAME	DNAME

8

M

The HR department needs to find the names and hire dates for all employees who were hired before their managers, along with their managers' names and hire dates.

Self Join with Outer Join

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

Query Builder

```
SELECT e.LAST_NAME AS "Employee",
       e.HIRE_DATE AS "Emp Hire Date",
       m.LAST_NAME AS "Manager",
       m.HIRE_DATE AS "Mgr Hire Date"
FROM HR.EMPLOYEES e
LEFT OUTER JOIN HR.EMPLOYEES m ON e.MANAGER_ID = m.EMPLOYEE_ID
WHERE e.HIRE_DATE < m.HIRE_DATE;
```

Script Output

Task completed in 0.164 seconds

Employee	Emp Hire	Manager	Mgr Hire
Garcia	13-01-11	King	17-06-13
Li	07-12-12	King	17-06-13
Kaufling	01-05-13	King	17-06-13
Gruenberg	17-08-12	Yang	21-09-15
Whalen	17-09-13	Yang	21-09-15
Jacobs	07-06-12	Yang	21-09-15
Brown	07-06-12	Yang	21-09-15
Higgins	07-06-12	Yang	21-09-15
Williams	25-06-15	James	03-01-16
Faviet	16-08-12	Gruenberg	17-08-12
Marlow	16-02-15	Frapp	10-04-15
Sarchand	27-01-14	Frapp	10-04-15
Bull	20-02-15	Frapp	10-04-15
Ladwig	14-07-13	Vollman	10-10-15
Bell	04-02-14	Vollman	10-10-15
Everett	03-03-15	Vollman	10-10-15
Rajs	17-10-13	Mourgos	16-11-17
Davies	29-01-15	Mourgos	16-11-17
Matos	15-03-16	Mourgos	16-11-17
Vargas	09-07-16	Mourgos	16-11-17
Walsh	24-04-16	Mourgos	16-11-17
Feeney	23-05-16	Mourgos	16-11-17

9

M

Display the EMPNO,ENAME,DNAME,LOC of those employees who are working as CLERK. Use the USING clause.

USING Clause

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

All Schemas
All Object Types
All Identifier Types
All Identifier Usages
Columns
Code
All Dependencies

Worksheet

Query Builder

```
SELECT e.EMPLOYEE_ID AS EMPNO,
       e.LAST_NAME AS ENAME,
       d.DEPARTMENT_NAME AS DNAME,
       l.CITY AS LOC
FROM HR.EMPLOYEES e
JOIN HR.DEPARTMENTS d USING (DEPARTMENT_ID)
JOIN HR.LOCATIONS l USING (LOCATION_ID)
WHERE e.JOB_ID LIKE '%CLERK';
```

Script Output

Task completed in 0.13500001 seconds

EMPNO	ENAME	DNAME	LOC
130	Atkinson	Shipping	South San Francisco
131	Marlow	Shipping	South San Francisco
132	Olson	Shipping	South San Francisco
133	Mallin	Shipping	South San Francisco
134	Rogers	Shipping	South San Francisco
135	Gee	Shipping	South San Francisco
136	Philtanker	Shipping	South San Francisco
137	Ladwig	Shipping	South San Francisco
138	Stiles	Shipping	South San Francisco
139	Seo	Shipping	South San Francisco
140	Patel	Shipping	South San Francisco
141	Rajs	Shipping	South San Francisco
199	Grant	Shipping	South San Francisco
143	Matos	Shipping	South San Francisco
144	Vargas	Shipping	South San Francisco
180	Taylor	Shipping	South San Francisco
181	Fleaur	Shipping	South San Francisco
182	Sullivan	Shipping	South San Francisco
183	Geoni	Shipping	South San Francisco
184	Sarchand	Shipping	South San Francisco
185	Bull	Shipping	South San Francisco
186	Dellinger	Shipping	South San Francisco

10

M

Display the ENAME,SAL,MGR,DNAME of employees whose salary is more than 2000. Use the ON clause.

On Clause

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

Oracle hr_install.sql 0.212 seconds

Worksheet Query Builder

```

SELECT e.LAST_NAME AS ENAME,
       e.SALARY AS SAL,
       e.MANAGER_ID AS MGR,
       d.DEPARTMENT_NAME AS DNAME
FROM HR.EMPLOYEES e
JOIN HR.DEPARTMENTS d ON (e.DEPARTMENT_ID = d.DEPARTMENT_ID)
WHERE e.SALARY > 2000;

```

Script Output

Task completed in 0.212 seconds

ENAME	SAL	MGR	DNAME
Whalen	4400	101	Administration
Martinez	13000	100	Marketing
Davis	6000	201	Marketing
Li	11000	100	Purchasing
Khoo	3100	114	Purchasing
Baida	2900	114	Purchasing
Tobias	2800	114	Purchasing
Himuro	2600	114	Purchasing
Colmenares	2500	114	Purchasing
Jacobs	6500	101	Human Resources
Weiss	8000	100	Shipping

ENAME	SAL	MGR	DNAME
Fripp	8200	100	Shipping
Kaufling	7900	100	Shipping
Vollman	6500	100	Shipping
Mourgos	5800	100	Shipping
Nayer	3200	120	Shipping
Mikkilineni	2700	120	Shipping
Landry	2400	120	Shipping
Markle	2200	120	Shipping
Bissot	3300	121	Shipping
Atkinson	2800	121	Shipping
Marlow	2500	121	Shipping

11

M

Display the EMPNO,ENAME,JOB,DEPTNO,DNAME,LOC of employees. Use LEFT OUTER JOIN.

LEFT OUTER JOIN

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

Worksheet

Query Builder

```

SELECT e.EMPLOYEE_ID AS EMPNO,
       e.LAST_NAME AS ENAME,
       e.JOB_ID AS JOB,
       e.DEPARTMENT_ID AS DEPTNO,
       d.DEPARTMENT_NAME AS DNAME,
       l.CITY AS LOC
FROM HR.EMPLOYEES e
LEFT OUTER JOIN HR.DEPARTMENTS d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
LEFT OUTER JOIN HR.LOCATIONS l ON d.LOCATION_ID = l.LOCATION_ID;

```

Script Output

Task completed in 0.208 seconds

EMPNO	ENAME	JOB	DEPTNO	DNAME	LOC
103	James	IT_PROG	60	IT	Southlake
104	Miller	IT_PROG	60	IT	Southlake
105	Williams	IT_PROG	60	IT	Southlake
106	Jackson	IT_PROG	60	IT	Southlake
107	Nguyen	IT_PROG	60	IT	Southlake
120	Weiss	ST_MAN	50	Shipping	South San Francisco
121	Fripp	ST_MAN	50	Shipping	South San Francisco
122	Kaufling	ST_MAN	50	Shipping	South San Francisco
123	Vollman	ST_MAN	50	Shipping	South San Francisco
124	Mourgos	ST_MAN	50	Shipping	South San Francisco
125	Nayer	ST_CLERK	50	Shipping	South San Francisco
126	Mikkilineni	ST_CLERK	50	Shipping	South San Francisco
127	Landry	ST_CLERK	50	Shipping	South San Francisco
128	Markle	ST_CLERK	50	Shipping	South San Francisco
129	Bissot	ST_CLERK	50	Shipping	South San Francisco
130	Atkinson	ST_CLERK	50	Shipping	South San Francisco
131	Marlow	ST_CLERK	50	Shipping	South San Francisco
132	Olson	ST_CLERK	50	Shipping	South San Francisco
133	Mallin	ST_CLERK	50	Shipping	South San Francisco
134	Rodgers	ST_CLERK	50	Shipping	South San Francisco

12

M

Display the ENAME,DNAME of employees. Use RIGHT OUTER JOIN.

RIGHT OUTER
JOIN

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

Worksheet

Query Builder

```

SELECT e.LAST_NAME AS ENAME,
       d.DEPARTMENT_NAME AS DNAME
FROM HR.EMPLOYEES e
RIGHT OUTER JOIN HR.DEPARTMENTS d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID;

```

Script Output

Task completed in 0.166 seconds

ENAME	DNAME
Whalen	Administration
Martinez	Marketing
Davis	Marketing
Baida	Purchasing
Tobias	Purchasing
Li	Purchasing
Khoo	Purchasing
Himuro	Purchasing
Colmenares	Purchasing
Jacobs	Human Resources
Bissot	Shipping
Dellinger	Shipping
Dilly	Shipping
Everett	Shipping
Feeney	Shipping
Fleaur	Shipping
Fripp	Shipping
Gee	Shipping
Geoni	Shipping
Grant	Shipping
Vollman	Shipping
Davies	Shipping

Oracle SQL Developer : Oracle

Find Database Object

Oracle

Go

Oracle hr_install.sql 0.153 seconds

Worksheet Query Builder

```

SELECT e.EMPLOYEE_ID AS EMPNO,
       d.DEPARTMENT_NAME AS DNAME,
       l.CITY AS LOC
FROM HR.EMPLOYEES e
FULL OUTER JOIN HR.DEPARTMENTS d ON e.DEPARTMENT_ID = d.DEPARTMENT_ID
LEFT OUTER JOIN HR.LOCATIONS l ON d.LOCATION_ID = l.LOCATION_ID;

```

Script Output x

Task completed in 0.153 seconds

EMPNO	DNAME	LOC
103	IT	Southlake
104	IT	Southlake
105	IT	Southlake
106	IT	Southlake
107	IT	Southlake
120	Shipping	South San Francisco
121	Shipping	South San Francisco
122	Shipping	South San Francisco
123	Shipping	South San Francisco
124	Shipping	South San Francisco
125	Shipping	South San Francisco
126	Shipping	South San Francisco
127	Shipping	South San Francisco
128	Shipping	South San Francisco
129	Shipping	South San Francisco
130	Shipping	South San Francisco
131	Shipping	South San Francisco
132	Shipping	South San Francisco
133	Shipping	South San Francisco
134	Shipping	South San Francisco
135	Shipping	South San Francisco
136	Shipping	South San Francisco